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ORIGINAL ARTICLE

Multi-institutional retrospective analysis of adverse events following rigid tracheobronchoscopy

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ABSTRACT

Background and objective: **Rigid tracheobronchoscopy (RTB) has seen an increasing interest over the last decades with the development of the field of IPM but no benchmark exists for complication rates in RTB. We aimed to establish benchmarks for complication rates in RTB.**
Methods: **A multicentric retrospective analysis of RTB performed between 2009 and 2015 in eight participating centres was performed.**
Results: **A total of 1546 RTB were performed over the study period. One hundred and thirty-one non-lethal complications occurred in 103 procedures (6.7%, 95% CI: 5.5–8.0%). The periprocedural mortality rate was 1.2% (95% CI: 0.6–1.8%). The 30-day mortality rate was 5.6% (95% CI: 4.5–6.8%). Complication rate increases further when procedures were performed in an emergency setting. Procedures in patients with MAO are associated with a higher 30-day mortality (8.1% vs 2.7%, $P < 0.01$) and a different complication profile when compared to procedures performed for BAS.**
Conclusion: **RTB is associated with a 6.7% non-lethal complication rate, a 1.2% periprocedural mortality rate and a 5.6% 30-day mortality in a large multicentre cohort of patients with benign and malignant airway disease.**

Key words: benign airway stenosis, bronchoscopy and interventional techniques, lung cancer, malignant airway obstruction, thoracic surgery.

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SUMMARY AT A GLANCE

RTB is associated with a 6.7% non-lethal complication rate, a 1.2% periprocedural mortality rate and a 5.6% 30-day mortality in a large multicentre cohort of patients with benign and malignant airway disease.

INTRODUCTION

Gustav Killian first described rigid tracheobronchoscopy (RTB) for therapeutic airway indications in 1895[1] and the technique was subsequently introduced in North America by Dr Chevalier Jackson.[2] RTB remained the only tool to access the airways until Shigeto Ikeda developed the first flexible bronchoscope in 1966[3] which has now replaced the rigid bronchoscope for the vast majority of bronchoscopic procedures. The development of the field of interventional pulmonary medicine (IPM) in the 1990s with the first dedicated airway stent by Jean-Francois Dumon[4] and the appearance on the market of self-expandable metallic stents dedicated for the airways has breathed new life into RTB. RTB is now used in the treatment of malignant airway obstructions (MAO), benign airway stenosis (BAS), foreign body removal, tracheobronchomalacia and haemoptysis to mention only the most frequent indications. RTB has become an important training objective in IPM and thoracic surgery training programmes but benchmarks for complication rates in RTB lack to allow quality monitoring in IPM programmes and help clinicians better understand the risks associated with RTB. We therefore decided to evaluate RTB complications in a retrospective multicentre study.

METHODS

An invitation to participate to this study was sent to 10 selected American Association for Bronchology and Interventional Pulmonology members, Canadian IPM training programme directors and international thoracic surgical programmes. Participants were sent a Microsoft Excel datasheet to collect patient, airway disease and procedure characteristics as well as procedural morbidity and mortality. All procedures performed between 1 January 2009 and 31 December 2015 at the participating centres were included. Procedures had to have been performed at least 3 months prior to chart review to be included. Subsequent procedures performed after the index case in an individual were included in the study if performed during the study period but excluded from mortality analysis. The institutional review board of the Centre Hospitalier Universitaire de Montreal approved this study (CRCHUM 14.074) and authorized a waiver of consent. Confidentiality of patient information was guaranteed.

All data entry fields were pre-defined (Appendix S1 in Supplementary Information). Major haemorrhage was defined as an intraprocedural haemorrhage of more than 200 mL or a haemorrhage requiring a major intervention (surgery, embolization or pulmonary isolation) or prolonged post-procedural endotracheal intubation secondary to the bleeding. Hypoxaemia was defined as a saturation of less than 90% for more than 1 min during the procedure. Arrhythmia was only recorded if it led to haemodynamic instability or required an immediate intervention. A procedure was considered emergent if the clinician judged it was imperative to proceed to the intervention within the following hours once the decision was made that the intervention was necessary or of potential benefit. Participating centres were asked to establish the indication of a procedure in cases with mixed indications based on the complaint that led to the procedure. Major haemorrhage, hypoxaemia, arrhythmia, myocardial infarction, haemodynamic instability, pneumothorax, pneumomediastinum, tracheal trauma, vocal cord injury, dental injury, labial injury, oesophageal injury and laryngeal oedema were grouped as non-lethal complications for statistical analysis.

Descriptive statistics were used to characterize patients (age, sex and American Society of Anesthesiologists (ASA) score), airway disease (lesion site and indication), procedures performed (duration, ventilation method, emergent status and technics used) and complications (non-lethal complications, unplanned intensive care unit (ICU) transfer, urgent reintervention, periprocedural mortality and 30-day mortality). Continuous and nominal variables were expressed using median with minimal–maximal values or percentage, respectively. Complications from procedures performed in MAO versus BAS, stent versus no stent, emergent versus non-emergent and index procedure in an individual versus subsequent procedures were compared using chi-square or Fisher's exact test. Correlations between volume of procedures performed in a centre and non-lethal complication rate, 30-day mortality rate and periprocedural mortality rate were expressed using the Pearson's correlation coefficients. The results were considered significant with P-values <0.05. All subsequent procedures in the same individual were excluded from mortality analysis. All analyses were conducted using the statistical package SAS, version 9.4 (SAS Institute Inc., Cary, NC, USA) and R (R Core Team (2016); Foundation for Statistical Computing, Vienna, Austria.).

RESULTS

A total of 1546 RTB procedures in 934 patients were performed in the eight participating institutions over the study period. The most frequent indications were MAO (n = 793, 51.3%) and BAS (n = 439, 28.4%) (Fig. 1). The most frequent location of lesion leading to the procedure was the trachea (n = 623, 40.3%) (Table 1). Two hundred and seventy-nine (18.0%)

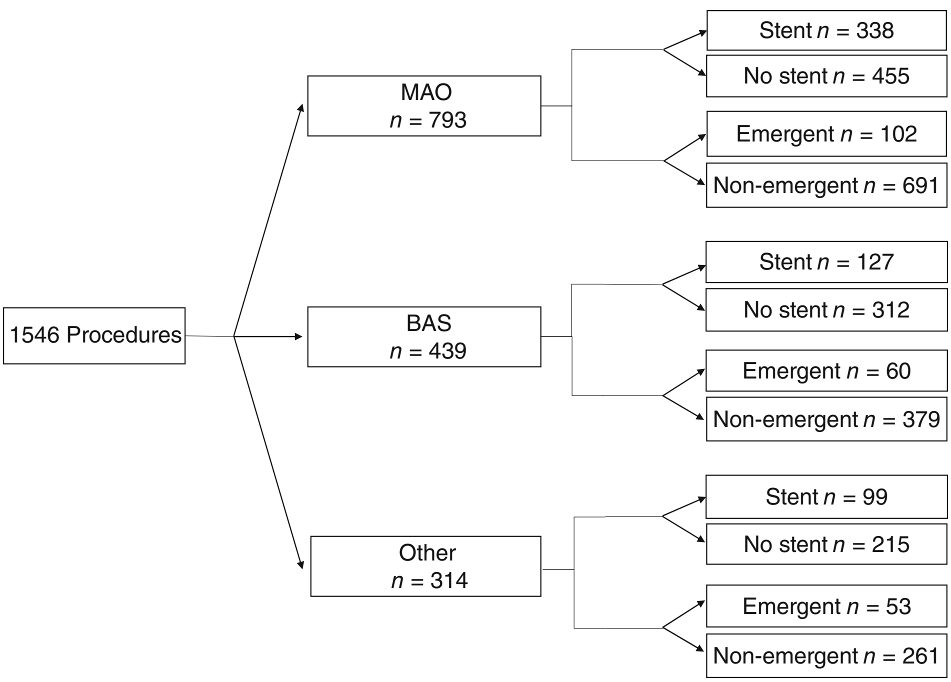

Figure 1 Procedure characteristics flow chart. MAO, malignant airway obstruction.

Table 1 Procedure characteristics

	All	MAO	BAS
Lesion site			
Trachea	623 (40.3)	245 (30.8)	212 (48.3)
BI	320 (20.7)	214 (27.0)	53 (12.1)
RMS	315 (20.4)	211 (26.6)	55 (12.5)
LMS	236 (15.3)	165 (20.8)	46 (10.5)
Lobar	198 (12.8)	132 (16.6)	13 (3.0)
Subglottic	107 (6.9)	11 (1.4)	95 (21.6)
Carina	55 (3.6)	44 (5.5)	1 (0.1)
Distal	16 (1.0)	5 (0.6)	5 (1.1)
Procedure performed			
Mechanical debulking or dilatation	643 (41.6)	441 (55.6)	160 (36.4)
Stent placement	564 (36.4)	338 (42.6)	127 (28.9)
Balloon dilatation	343 (22.2)	184 (23.2)	142 (32.3)
Stent removal	261 (16.9)	131 (16.5)	56 (12.8)
Laser	215 (13.9)	68 (8.6)	60 (13.7)
APC	61 (3.9)	52 (6.6)	0 (0)
Electrocautery	13 (0.8)	10 (1.3)	0 (0)
Cryotherapy	12 (0.8)	9 (1.1)	1 (0.2)
Other[†]	302 (19.5)	139 (17.5)	66 (15.0)

[†]Other includes foreign body retrieval, microdebrider use, haemostatic control, emphysema valve insertion and procedures not listed in the table or in this list.

APC, argon plasma coagulation; BAS, benign airway stenosis; BI, bronchus intermedius; LMS, left mainstem bronchus; MAO, malignant airway obstruction; RMS, right mainstem bronchus.

procedures involved only airways distal to the mainstem bronchi. The most frequent procedures performed were mechanical debulking or dilatation using the rigid bronchoscope (n = 643, 41.6%) and airway stent placement (n = 564, 36.5%). Two hundred and fifteen (14.1%) procedures were emergent. Median procedure duration was 34 min (range: 1–455 min). Jet ventilation was used in the majority of cases (n = 1136, 73.5%). Median length of stay was 1 day (range: 0–138 days) with 192 (12%) procedures not requiring an overnight stay.

One hundred and thirty-one non-lethal complications occurred in 103 (6.7%, 95% CI: 5.5–8.0%) procedures. Sixty (3.9%) and 57 (3.7%) procedures led to unplanned ICU transfer and urgent reintervention, respectively. The periprocedural mortality rate was 1.2% (n = 18, 95% CI: 0.6–1.8%). The 30-day mortality rate was 5.6% (n = 86, 95% CI: 4.5–6.8%).

Sixty-six non-lethal complications occurred in 53 (6.7%, 95% CI: 5.1–8.7%) RTB for MAO. The most frequent complications were major haemorrhage (n = 18, 2.3%), haemodynamic instability (n = 15, 1.9%) and hypoxaemia (n = 12, 1.5%). Thirty-seven (4.7%) and 23 (2.9%) procedures led to unplanned ICU transfer and urgent reintervention, respectively. The periprocedural mortality rate was 1.4% (95% CI: 0.6–3.0%). The 30-day mortality rate was 8.1% (95% CI: 5.9–10.9%).

Table 2 Complications in MAO versus BAS

	MAO	BAS	*P*-value
Non-lethal complications	53 (6.7)	22 (5.0)	0.26
Major haemorrhage	18 (2.3)	1 (0.2)	
Hypoxaemia	12 (1.5)	5 (1.1)	
HD instability	15 (1.9)	2 (0.5)	
Labial injury	6 (0.8)	6 (1.4)	
Dental injury	2 (0.3)	3 (0.7)	
Vocal cord injury	2 (0.3)	2 (0.5)	
Laryngeal oedema	2 (0.3)	3 (0.7)	
Arrythmia	4 (0.5)	1 (0.2)	
MI	1 (0.1)	2 (0.5)	
Pneumothorax	3 (0.4)	1 (0.2)	
Other	1 (0.1)	1 (0.3)	
Urgent reintervention	23 (2.9)	18 (4.1)	0.32
Unplanned ICU transfer	37 (4.7)	10 (2.3)	0.04
Periprocedural mortality	7 (1.4)	1 (0.5)	0.44
30-Day mortality	41 (8.1)	6 (2.7)	<0.01

BAS, benign airway stenosis; HD, haemodynamic instability; ICU, intensive care unit; MAO, malignant airway obstruction; MI, myocardial infarction.

Table 3 Complications in procedures with stent placement versus no stent placement

	Stent	No stent	*P*-value
Non-lethal complications	44 (7.8)	59 (6.0)	0.20
Major haemorrhage	5 (0.9)	21 (2.1)	
Hypoxaemia	11 (2.0)	11 (1.1)	
HD instability	10 (1.8)	18 (1.8)	
Labial injury	8 (1.4)	6 (0.6)	
Dental injury	5 (0.9)	2 (0.2)	
Vocal cord injury	2 (0.4)	3 (0.3)	
Laryngeal oedema	3 (0.5)	4 (0.4)	
Arrythmia	4 (0.7)	5 (0.5)	
MI	2 (0.4)	1 (0.1)	
Pneumothorax	3 (0.5)	3 (0.3)	
Other	3 (0.5)	1 (0.1)	
Urgent reintervention	37 (6.6)	20 (2.0)	<0.01
Unplanned ICU transfer	23 (4.1)	37 (3.8)	0.79
Periprocedural mortality	8 (2.1)	4 (0.7)	0.07
30-Day mortality	39 (10.4)	28 (4.9)	<0.01

HD, haemodynamic; ICU, intensive care unit; MI, myocardial infarction.

Twenty-seven non-lethal complications occurred in 22 (5.0%, 95% CI: 3.2–7.5%) RTB for BAS. The most frequent complications were labial injury (n = 6, 1.4%), hypoxaemia (n = 5, 1.1%), dental injury (n = 3, 0.7%) and laryngeal oedema (n = 3, 0.7%). Ten (2.3%) and 18 (4.1%) procedures led to unplanned ICU transfer and urgent reintervention, respectively. The periprocedural mortality rate was 0.5% (95% CI: 0–2.5%). The 30-day mortality rate was 2.7% (95% CI: 1.0–5.8%).

Procedures performed for MAO when compared to procedures performed for BAS were associated with

Table 4 Complications in emergent versus non-emergent procedures

	Emergent	Non-emergent	P-value
Non-lethal complications	34 (15.8)	68 (5.2)	<0.01
Major haemorrhage	10 (4.7)	16 (1.2)	
Hypoxaemia	7 (3.3)	14 (1.1)	
HD instability	8 (3.7)	20 (1.5)	
Labial injury	8 (3.7)	6 (0.5)	
Dental injury	4 (1.9)	3 (0.2)	
Vocal cord injury	0 (0)	4 (0.3)	
Laryngeal oedema	1 (0.5)	6 (0.5)	
Arrythmia	2 (0.9)	7 (0.5)	
MI	1 (0.5)	2 (0.2)	
Pneumothorax	1 (0.5)	5 (0.4)	
Other	1 (0.1)	3 (0.4)	
Urgent reintervention	30 (13.9)	27(2.1)	<0.01
Unplanned ICU transfer	22 (10.2)	37 (2.8)	<0.01
Periprocedural mortality	4 (2.8)	8 (1.0)	0.09
30-Day mortality	22 (15.5)	45 (5.7)	<0.01

HD, haemodynamic; ICU, intensive care unit; MI, myocardial infarction.

more unplanned ICU transfers (n = 37/793 vs 10/439, 4.7% vs 2.3%, P = 0.043) and 30-day mortality (n = 41/504 vs 6/220, 8.1% vs 2.7%, P = 0.005) (Table 2).

Airway stent placement was associated with higher rates of urgent reintervention (n = 37/564 vs 20/982, 6.6% vs 2.0%, P < 0.001) and 30-day mortality (n = 39/376 vs 28/567, 10.4% vs 4.9%, P = 0.002) (Table 3).

Emergent procedures were associated with more non-lethal complications (n = 34/215 vs 68/1314, 15.8% vs 5.2%, P < 0.001), unplanned ICU transfers (n = 22/215 vs 37/1313, 10.2% vs 2.8%, P < 0.001), urgent reinterventions (n = 30/215 vs 27/1312, 14.0% vs 2.1%, P < 0.0001) and 30-day mortality (n = 22/142 vs 45/792, 15.5% vs 5.7%, P = 0.001) (Table 4).

Subsequent procedures in an individual after an index procedure were associated with lower rates of non-lethal complications (n = 27/603 vs 76/943, 4.5% vs 8.1%, P = 0.0063) while urgent reintervention (n = 23/603 vs 34/941, 3.8% vs 3.6%, P = 0.8902) and unplanned ICU transfer (n = 19/603 vs 41/943, 3.2% vs 4.4%, P = 0.2805) rates were similar.

The correlation between the volume of procedures performed in a centre and 30-day mortality (R = −0.36, P = 0.359) as well as non-lethal complication rate (R = −0.43, P = 0.285) were low while the correlation was higher for periprocedural mortality (R = −0.69, P = 0.19).

DISCUSSION

This large retrospective multicentre study is the largest reported series examining RTB complication rates. It establishes benchmarks for complication rates and demonstrates the risks associated with RTB. In the overall population, non-lethal complication rate, periprocedural mortality rate and 30-day mortality rate were 6.7% (95% CI: 5.5–8.0%), 1.2% (95% CI: 0.6–1.8%) and 5.6% (95% CI: 4.5–6.8%), respectively. Procedures performed for MAO when compared to procedures performed for BAS were associated with higher rates of unplanned ICU transfer (4.7% vs 2.3%) and 30-day mortality (8.1% vs 2.7%). Non-lethal complication rates did not differ (6.7% vs 5.0%), although types of complication differed with major haemorrhage (2.3%) and haemodynamic instability (1.5%) composing the majority of complications in MAO while labial (1.4%) and dental injury (0.7%) as well as laryngeal oedema (0.7%) composed the majority of complications in BAS. These benchmarks may be used in the future by clinicians to evaluate their local complication rates and to inform patients adequately of potential complications when obtaining informed consent for RTB. Unfortunately, this study did not encompass markers to objectively explore the correlation between the frailty of patients and complication rates. The initial data collection worksheet for the study included obtaining the ASA score for every patient; however, the information was not available in the majority of cases as different centres utilized different scores. The largely higher complication rate in patients undergoing urgent procedures may reflect this relationship. The complication rates obtained may also vary with the complexity of the performed procedure; however, this hypothesis could not be explored as no tool exists to objectively evaluate the complexity of a procedure.

Few studies have focused on the complications of RTB.[5–9] Large studies such as the AQuIRE cohort[5] have indirectly reported complication rate for RTB. This large prospective database of IPM procedures performed in the United States included a cohort of therapeutic bronchoscopies performed for MAO, two-thirds of which were RTB (733 RTB). Complications were reported in 3.9% of cases overall with a procedural mortality rate of 0.5% and a 30-day mortality of 14.8%. Complication rates reported in this study vary from centre to centre and range from 0.9% to 11.7%. This study focused on outcomes and very little information was recorded about adverse events. Differences in the recorded complications may also explain the discrepancies in results. It is also important to note that one-third of procedures were performed without RTB, which may reflect a decreased complexity of procedure. Our results add to Ost *et al.*'s[5] finding by providing far more details about complications in an international cohort limited to RTB and not limited to MAO. Ernst *et al.*[6] also published a cohort of therapeutic bronchoscopic procedures which included 483 RTB in benign and malignant conditions with a more important focus on complications. They reported a 19.8% complication rate and 7.8% 30-day mortality. These results in a population which included benign airway disease and a vast majority of interventions by RTB are in line with ours when considering minor bleeding, which was not reported in our cohort (total complication rate: 22.1% vs 19.8%). We elected not to report minor bleeding as we felt retrospective reporting of such a complication may be unreliable and this complication is typically not clinically relevant. Our results add to Ernst *et al.*'s findings by validating them in a fourfold larger cohort but also by including smaller volume centres and non-American centres.

Although large and multicentric, this study's main weakness is its retrospective nature. Certain complications are not as reliably identified on chart review as others. Major complications such as periprocedural mortality, major haemorrhage or ICU transfer are well documented in medical charts; however, minor complications such as transient hypoxia may not be as well documented. We feel that complications with a clinical repercussion are well documented in medical charts. Complications that were not documented may be of less interest to inform a patient properly of potential complications or for a clinician to properly perform quality control audits. The retrospective nature of this study also kept us from obtaining reliable information on the clinical response to the procedures performed as there was no standardized follow-up with objective measures available.

A multivariate analysis would have been of interest to clarify the higher complication rates associated with stent insertion; however, it could not be reliably performed due to the low number of events.

Correlations between the volume of procedure performed at a centre and the non-lethal complication rate as well as 30-day mortality were not as strong as expected. Correlation with periprocedural mortality was stronger. Further data are required to clarify these correlations. The relatively small number of centres included in this study and the relative homogeneity of the participating centres may have prevented us from truly exploring the correlation between volume and complications. Six of the eight centres performed between 20 and 40 RTB annually with only two outlying centres, respectively, performing 7 and 117 RTB annually. We also did not account for the number of different operators per site and the operators' cumulative experience. As an example, a small-volume centre with only one operator with a large experience in RTB accumulated over years may be equivalent to a larger volume centre with more junior operators sharing the volume. One could also argue that the correlation found between volume and complications in our cohort would be reproducible in a study including more centres with a wider array of volumes. Our complication rate being in line with Ernst *et al.*[6] whose work included larger volume centres (average of 92.3 RTB annually) is an argument in favour of this hypothesis. A weaker correlation could be explained by the fact that a low procedure volume may be associated with more stringent patient selection, with centres with less experience selecting less complex procedures in patients who are less physically perturbed or with less complex anatomical airway problems (carinal, severe stenosis, severe hypoxaemia, etc.).

In conclusion, RTB is associated with a 6.7% non-lethal complication rate, a 5.6% 30-day mortality and a 1.2% periprocedural mortality rate in a large multicentre cohort of patients with benign and malignant airway disease. Complication rates increase further when procedures are performed in an emergency setting. Further research with objective outcomes and prolonged follow-up is needed to obtain benchmarks for long-term outcomes.

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Abbreviations: ASA, American Society of Anesthesiologists; BAS, benign airway stenosis; HD, haemodynamic; ICU, intensive care unit; IPM, interventional pulmonary medicine; MAO, malignant airway obstruction; MI, myocardial infarction; RTB, rigid tracheobronchoscopy

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Supplementary Information

Additional supplementary information can be accessed via the *html* version of this article at the publisher's website.

Appendix S1. Definitions and collected parameters.